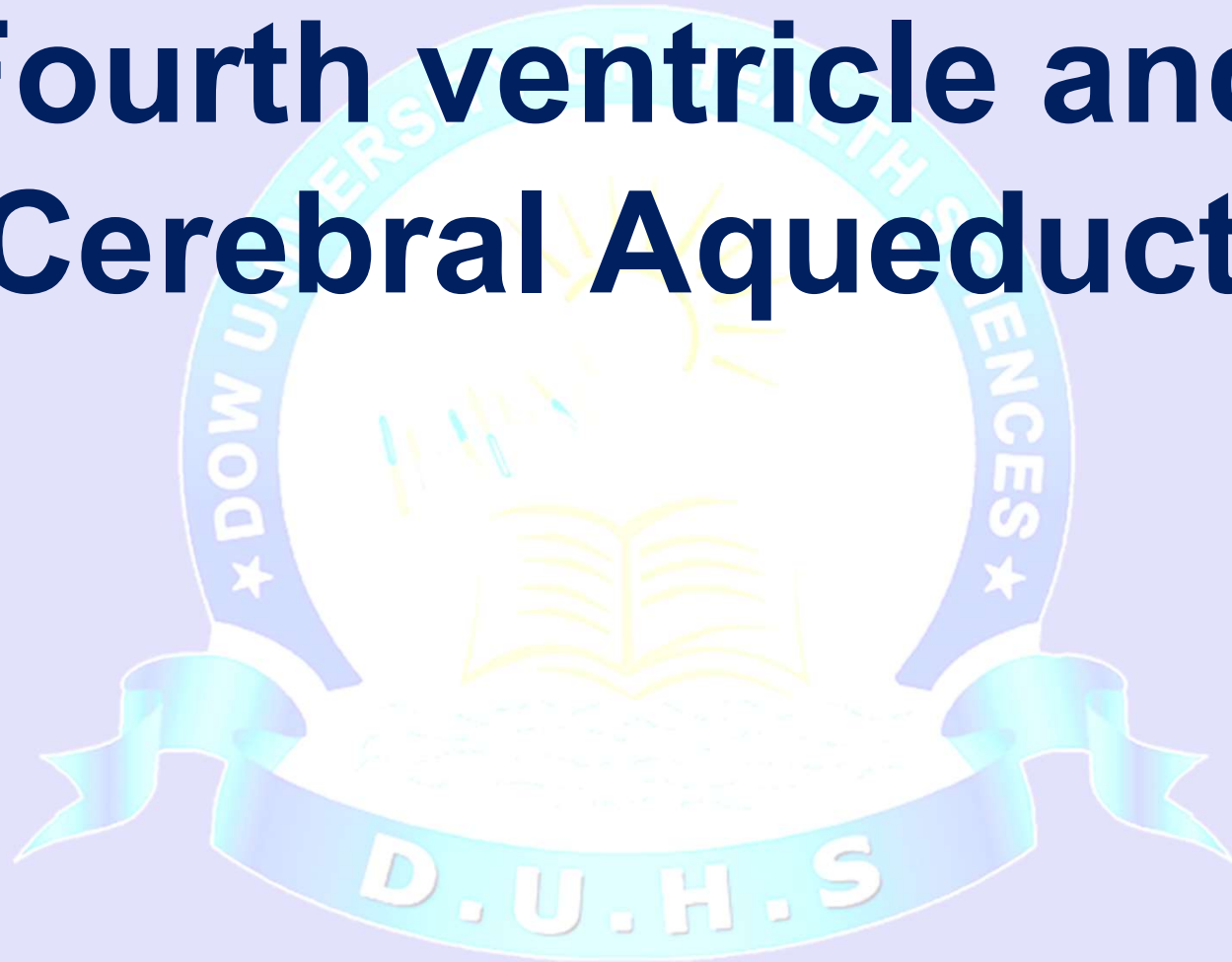


Fourth ventricle and Cerebral Aqueduct



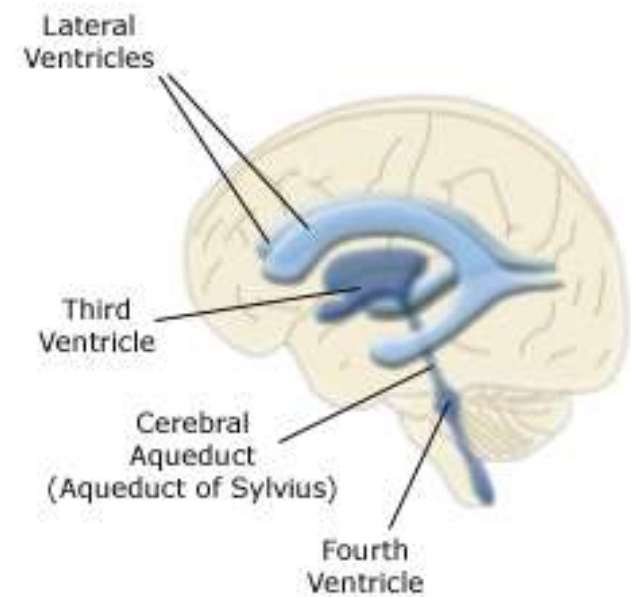
LECTURE OUTLINE

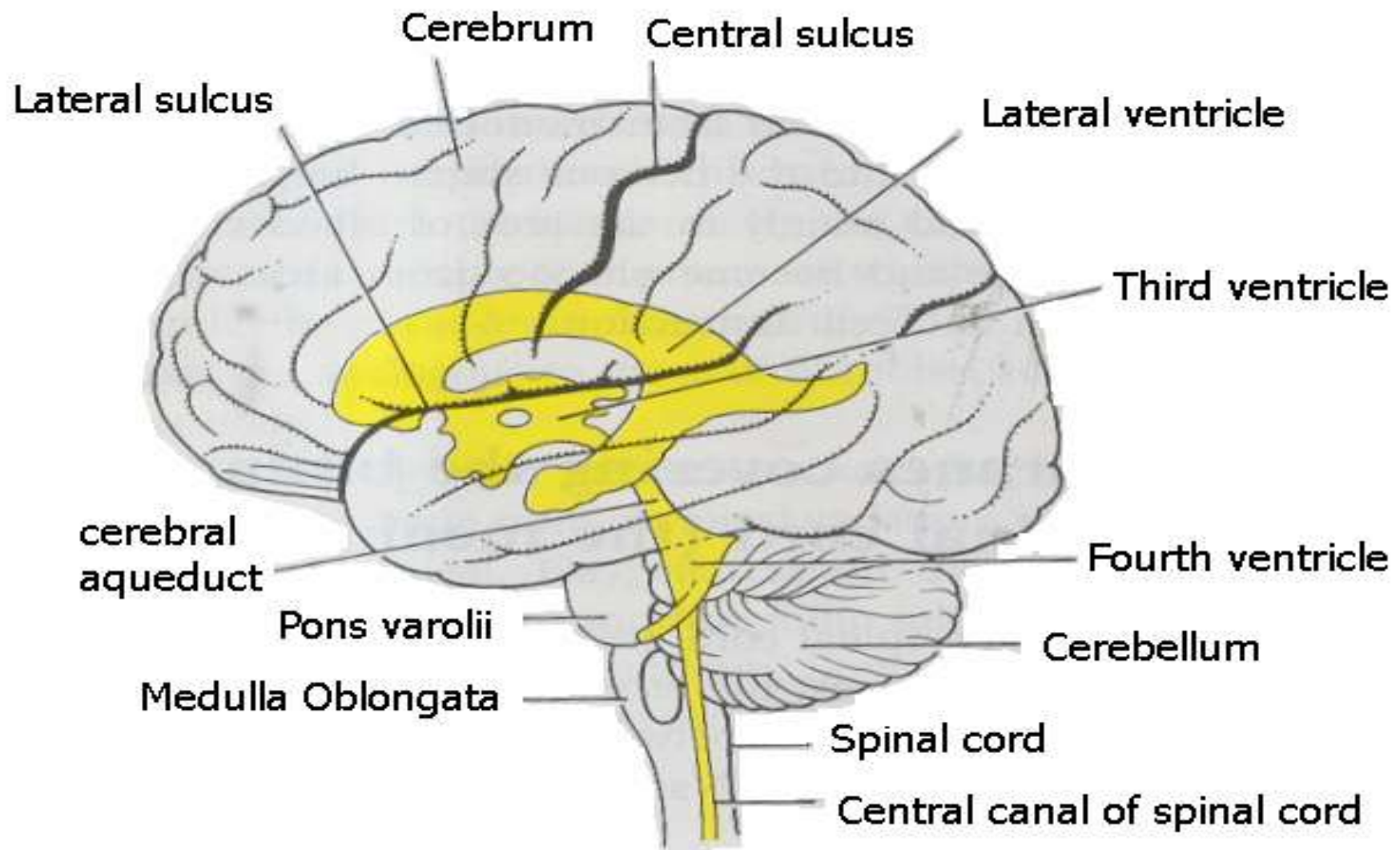
At the end of the lecture ,student should be able to:

- Name the ventricles of brain along with their location.
- Explain the structure and location of fourth ventricle
- Explain the structure and location of cerebral aqueduct
- Know the normal CSF secretion and circulation.
- Define the Blood brain barrier
- Discuss the Applied anatomy of ventricles and CSF flow

VENTRICLES OF BRAIN

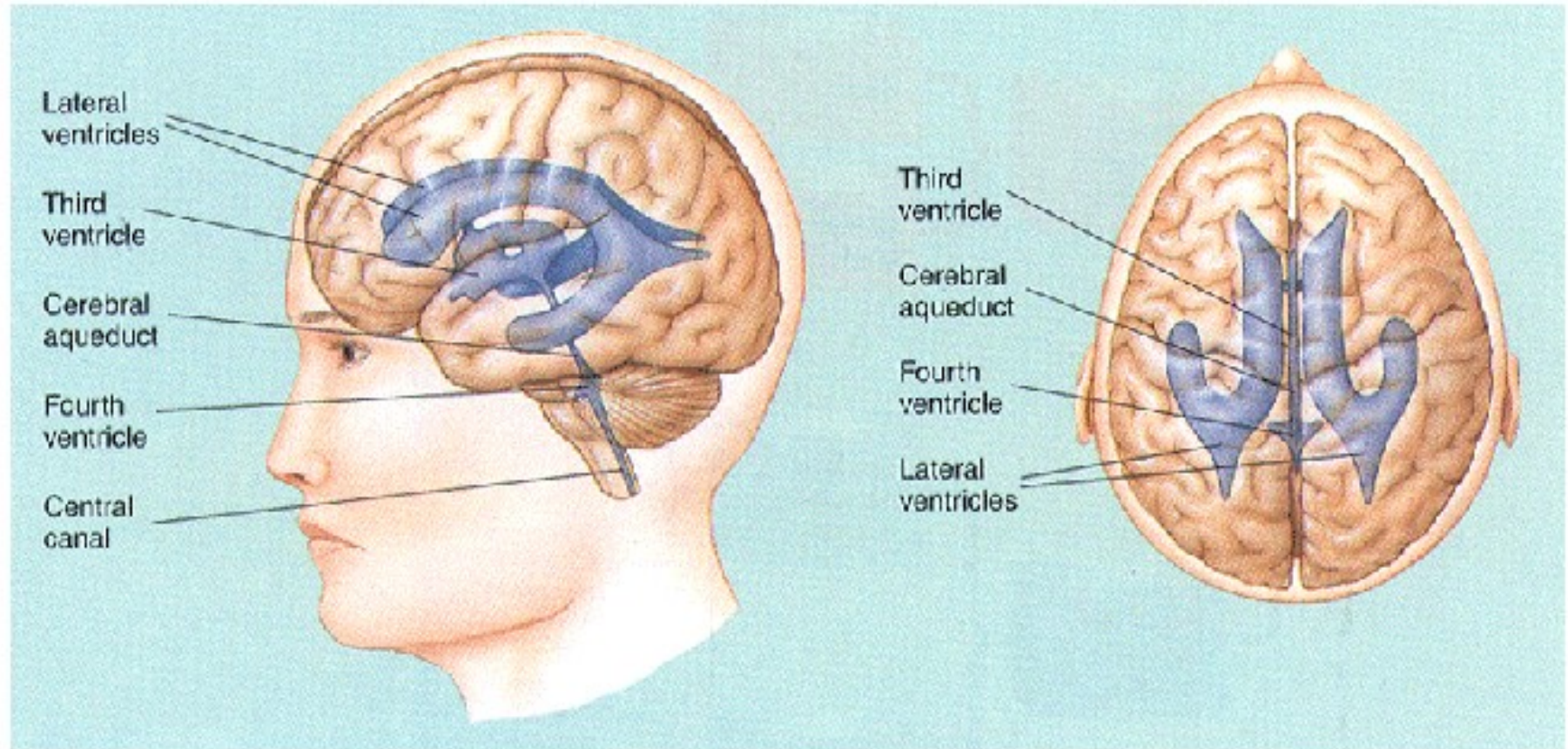
- They are cavities within the brain
- Irregularly shaped
- Lined with ependymal cells
- Four ventricles
- **Lateral ventricle** (each hemisphere)
- **Third ventricle** (in diencephalon)
- **Cerebral aqueduct** (in midbrain)
- **Fourth ventricle**: (lies between pons, upper part of medulla and cerebellum continuous with central canal of spinal cord)



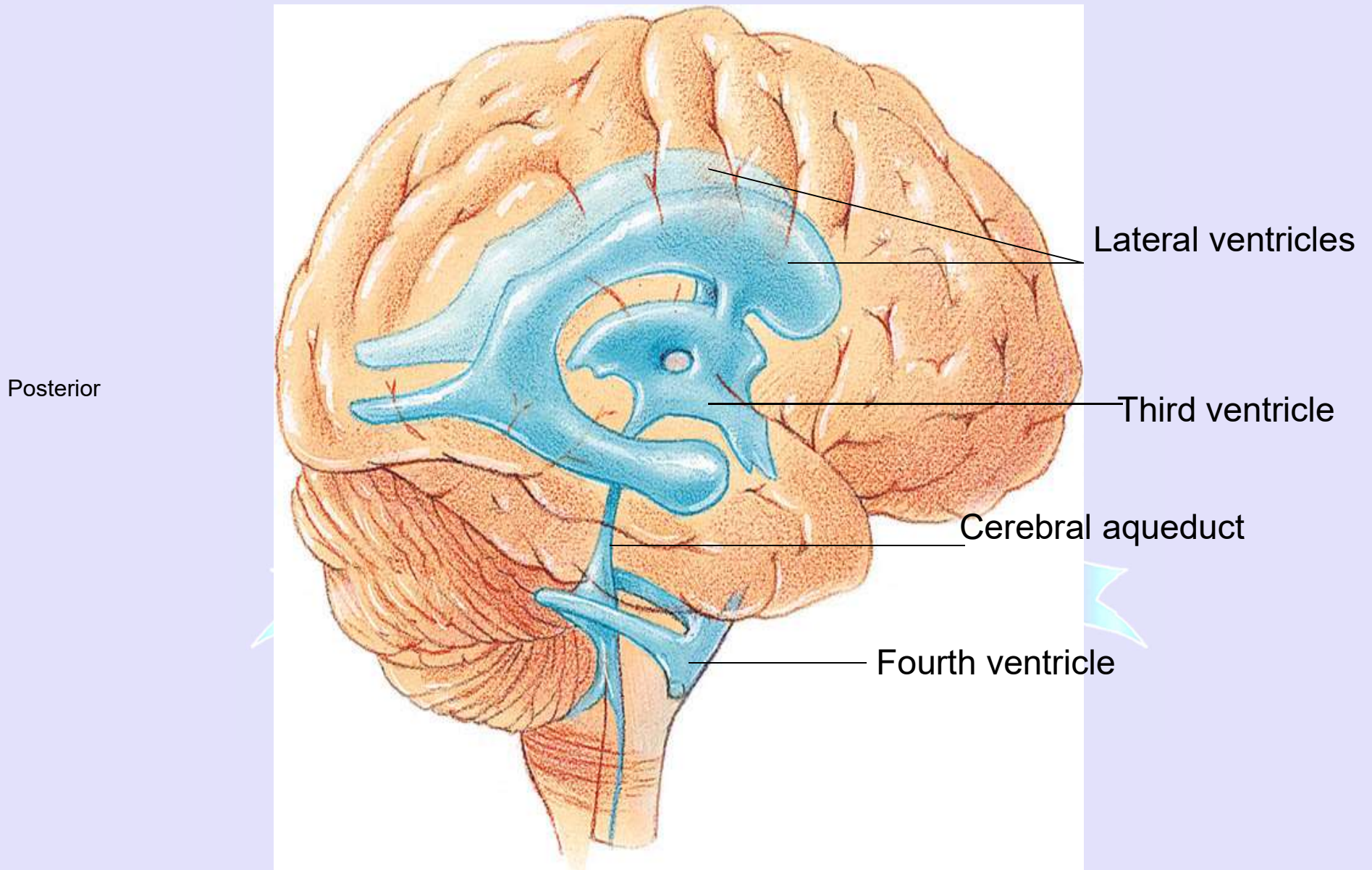


The positions of the ventricles of the brain (in yellow) superimposed on its surface. Viewed from the left side.

► Cerebral Ventricles



VENTRICLES OF BRAIN



CEREBROSPINAL FLUID

Surrounds and bathes the CNS

- Functions:
 1. Supporting of brain and spinal cord
 2. Transport of nutrients, chemical messengers, and waste products

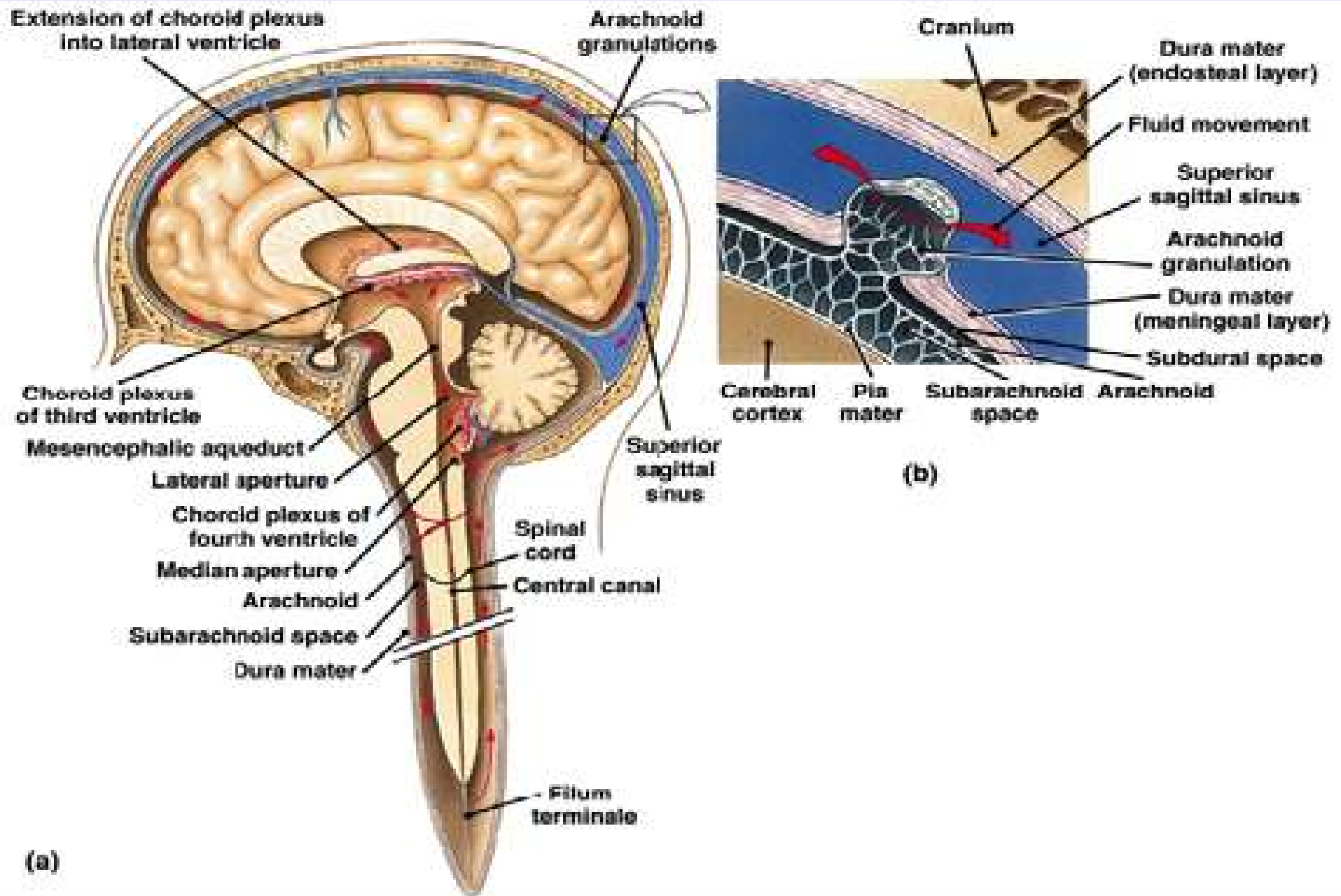
THE FORMATION OF CSF

- Choroid plexus: contains specialized ependymal cells and capillaries (**500ml/day**), **total volume: 150ml**
- Choroid plexuses secrete CSF into ventricles

Circulation:

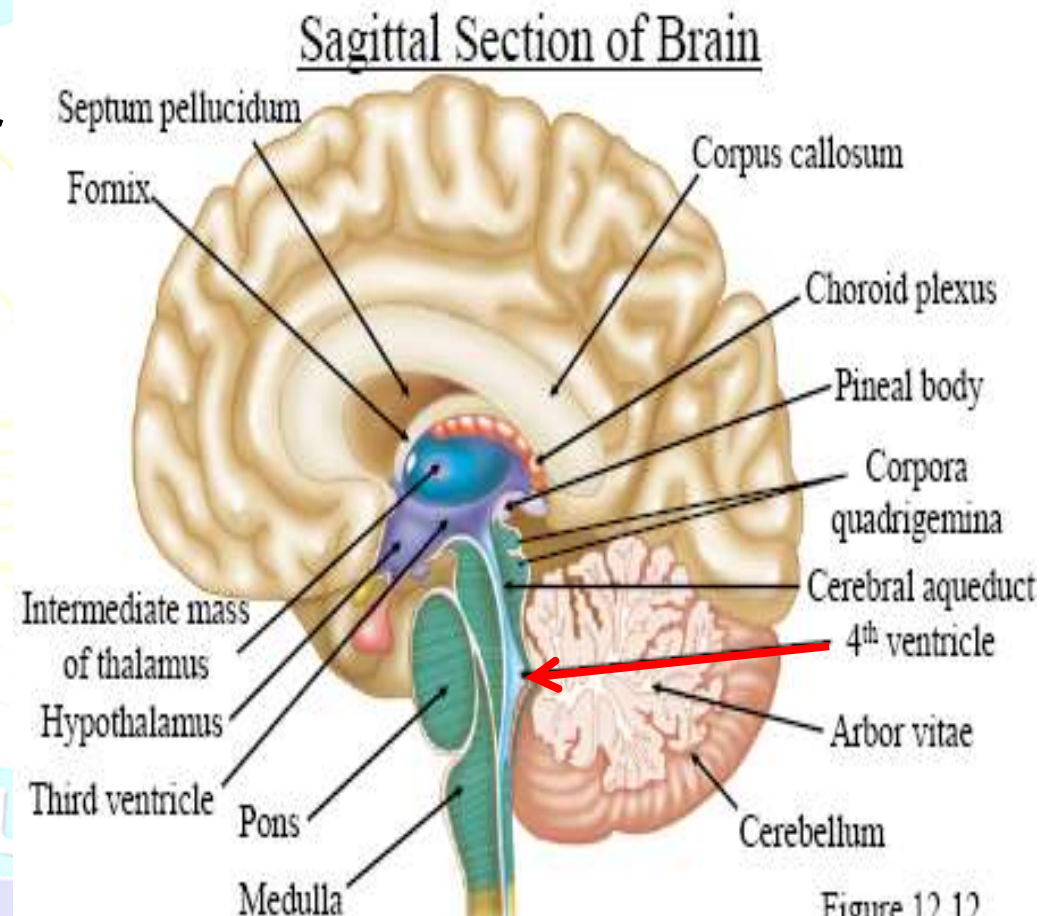
- from choroid plexus to ventricles and central canal of spinal cord to subarachnoid space to sinuses
- CSF reaches subarachnoid space through two lateral apertures and single medial aperture in the 4th ventricle
- Arachnoid villi (granulations): penetrate dura mater meningeal layer of venous sinuses, CSF absorbed into the venous circulation

SECRETION AND CIRCULATION OF CSF



FOURTH VENTRICLE

- The fourth ventricle is a cavity which lies posterior to the pons and upper half of the medulla oblongata and anterior to the cerebellum.
- It is continuous with the cerebral aqueduct (mesencephalic or duct of Sylvius) above and the central canal of the spinal cord in the lower half of the medulla.
- On each side, a narrow prolongation, the lateral recess, projects around the brainstem; its lateral aperture (forame of Luschka) lies below the cerebellar flocculus.

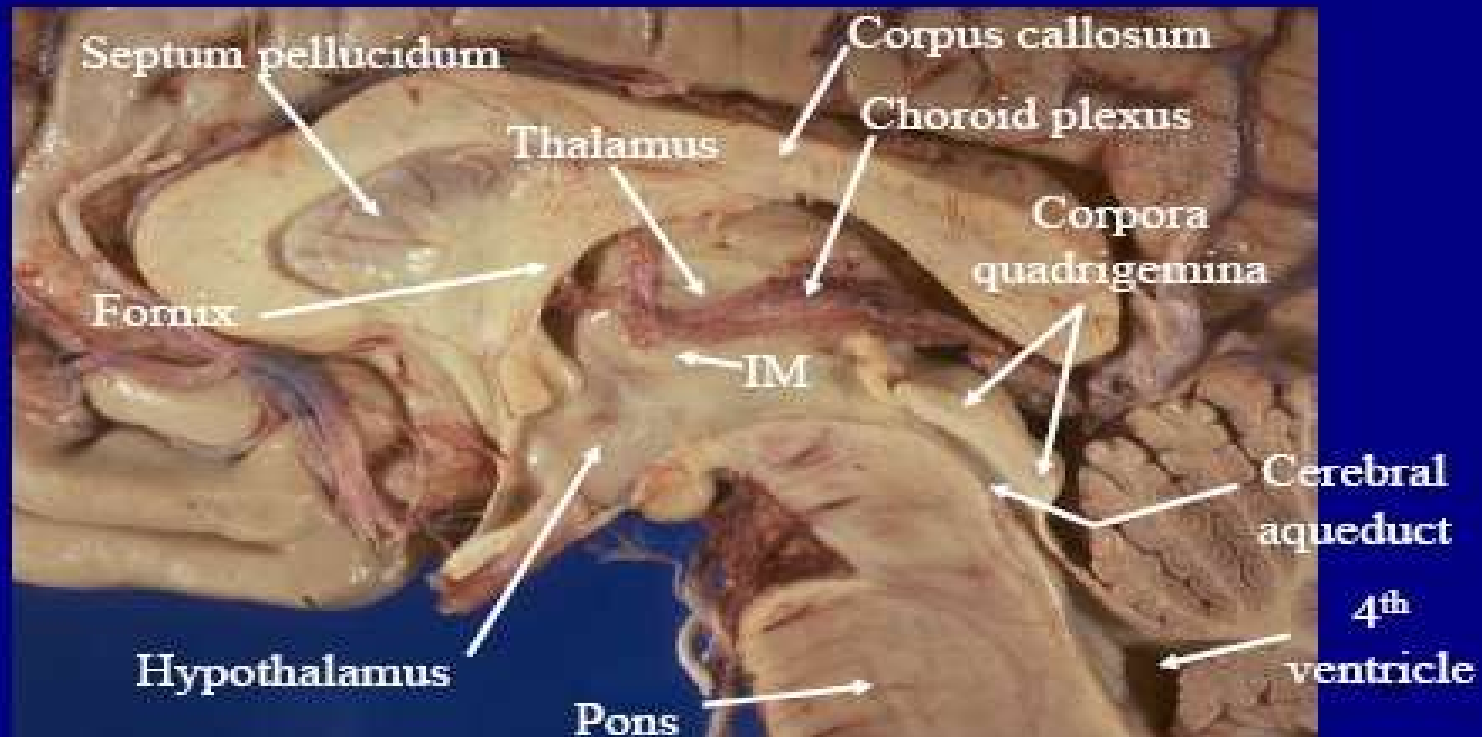


FOURTH VENTRICLE

The fourth ventricle has

- lateral boundaries,
- a roof
- and a floor.

Sagittal Section of Midbrain and Diencephalon



FOURTH VENTRICLE

The lateral boundaries

- are formed on each side by the superior cerebellar peduncle,
- the inferior cerebellar peduncle
- and the cuneate and gracile tubercles.

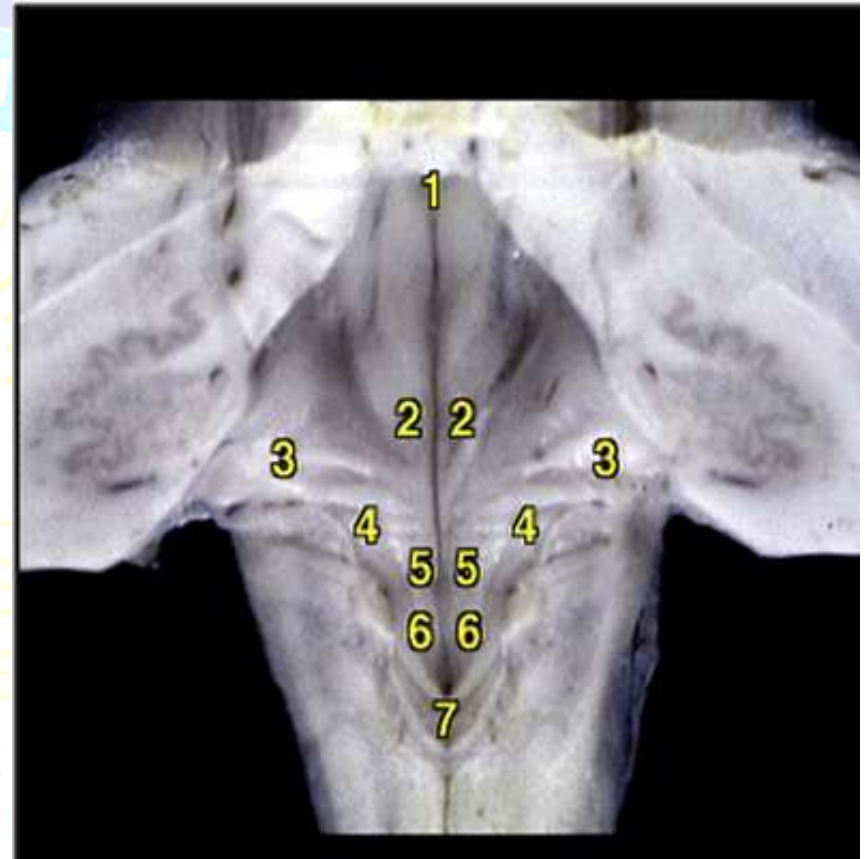
Roof of the fourth ventricle

- Formed by thin laminae of white matter.
- The lower has a median aperture (foramen of Magendie);
- cerebrospinal fluid escapes through this opening and lateral apertures into the subarachnoid space
- their blockage can produce one type of hydrocephalus.

FOURTH VENTRICLE

The floor of the fourth ventricle

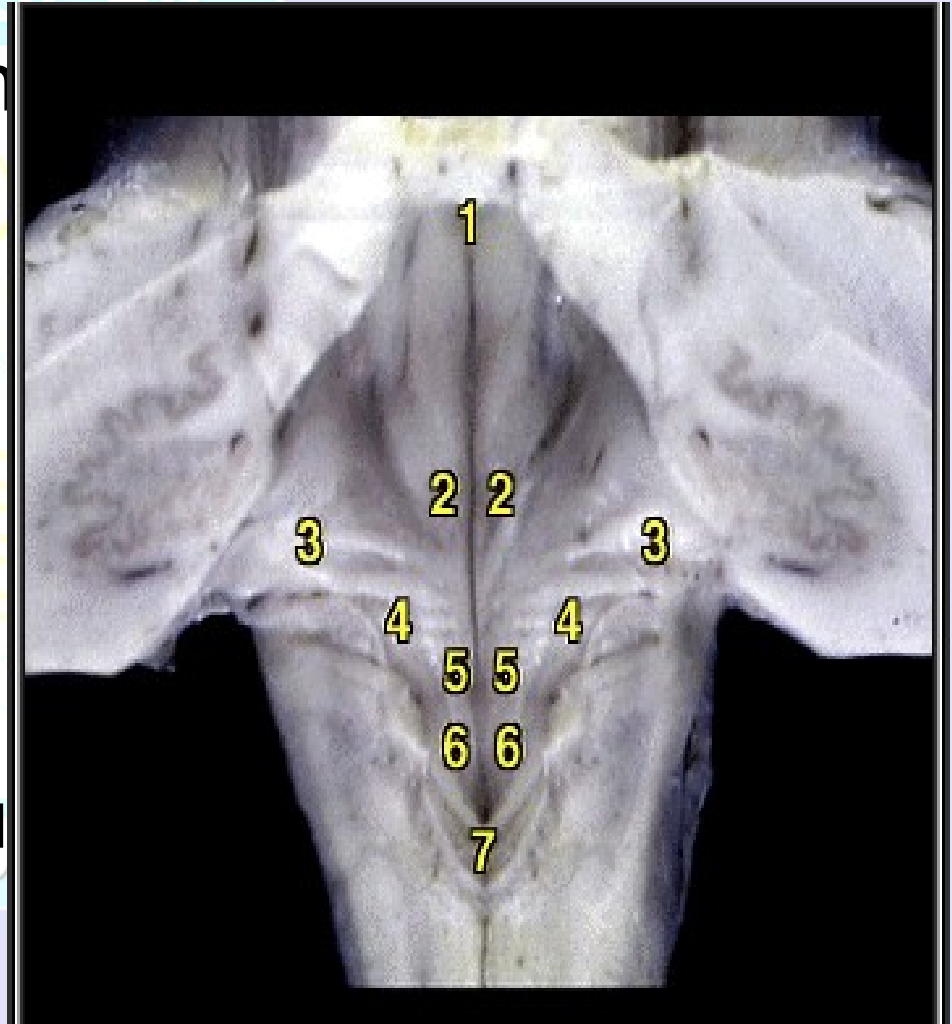
- also known as rhomboid fossa,
is formed by the dorsal surfaces of the pons and medulla oblongata.



- 1Median sulcus
- 2Superior cerebellar peduncle
- 3Dentate nucleus
- 4Floor of fourth ventricle

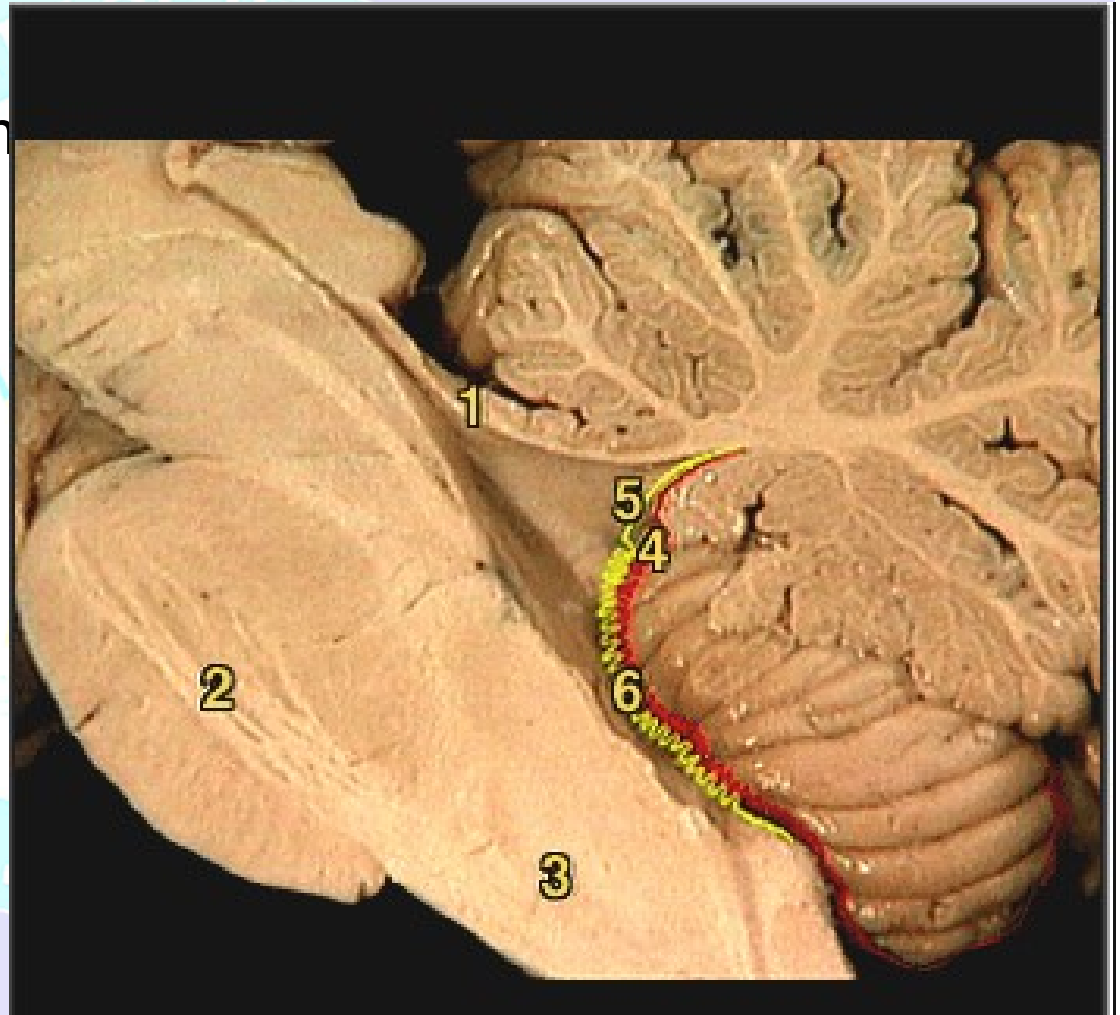
Floor of Fourth Ventricle, Posterior View

1. Substantia ferruginea
2. Facial colliculus
3. Striae medullares
4. Vestibular area
5. Hypoglossal triangle
6. Vagal triangle
7. Calamus scriptorius



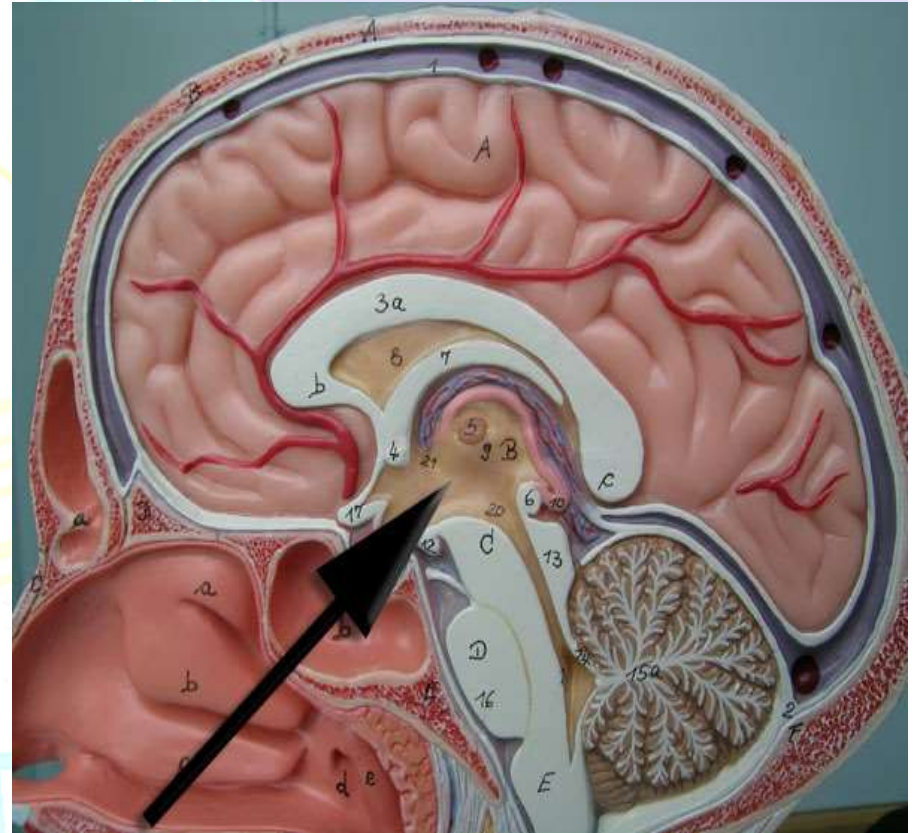
Fourth Ventricle, Sagittal Section, Medial View

1. Superior medullary velum
2. Pons
3. Medulla oblongata
4. Pia mater
5. Ependyma
6. Choroid plexus



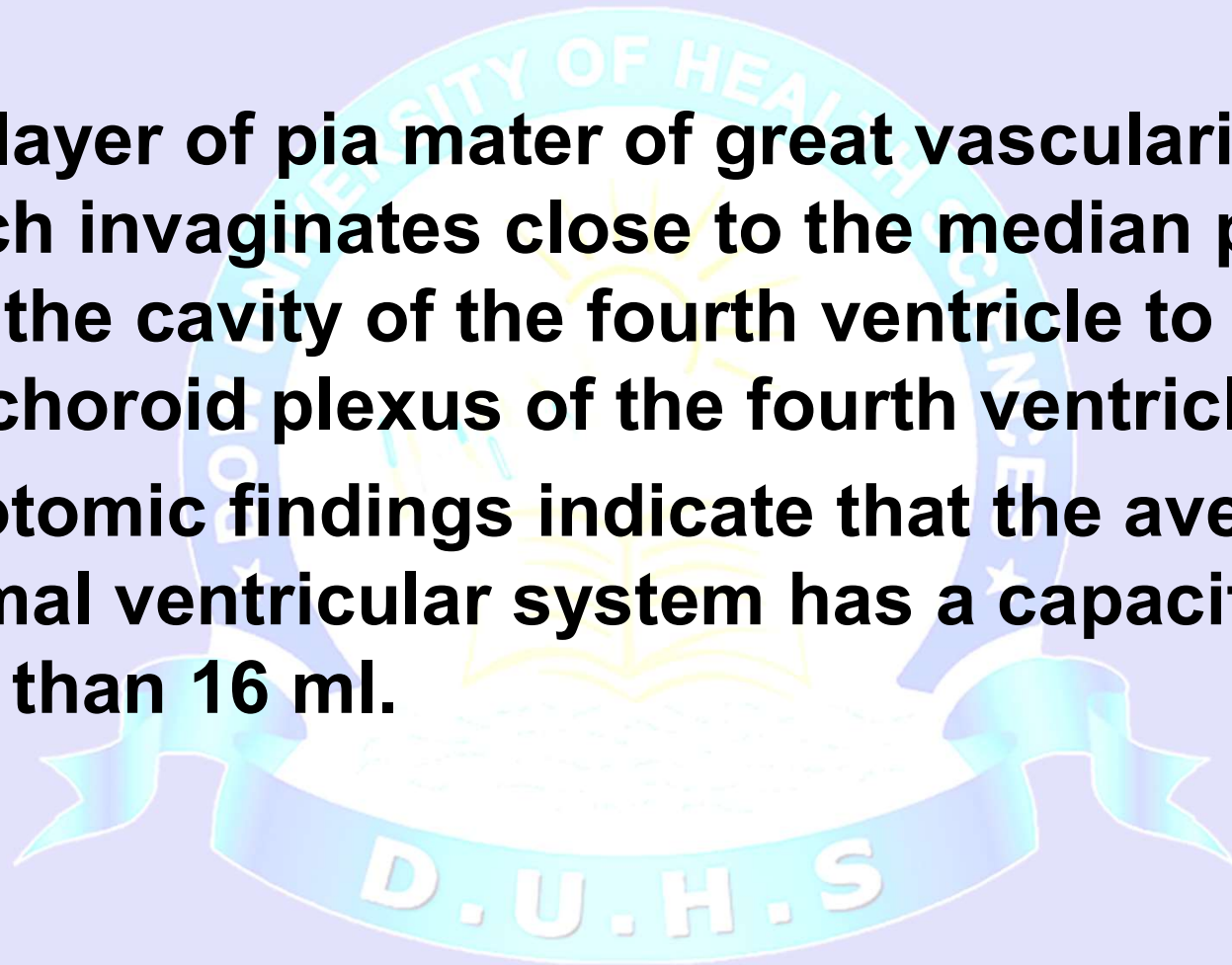
CEREBRAL AQUEDUCT

- The cerebral aqueduct is a narrow canal in the midline
- Connects the third and fourth ventricle.
- It is 1.5 cm long and 1-2 mm, in diameter.
- Its floor is formed by the tegmentum of the midbrain.
- Its roof consists of the quadrigeminal body of the midbrain and posterior commissure.



TELA CHOROIDEA

- is a layer of pia mater of great vascularity which invaginates close to the median plane into the cavity of the fourth ventricle to form the choroid plexus of the fourth ventricle.
- Anatomic findings indicate that the average normal ventricular system has a capacity of less than 16 ml.

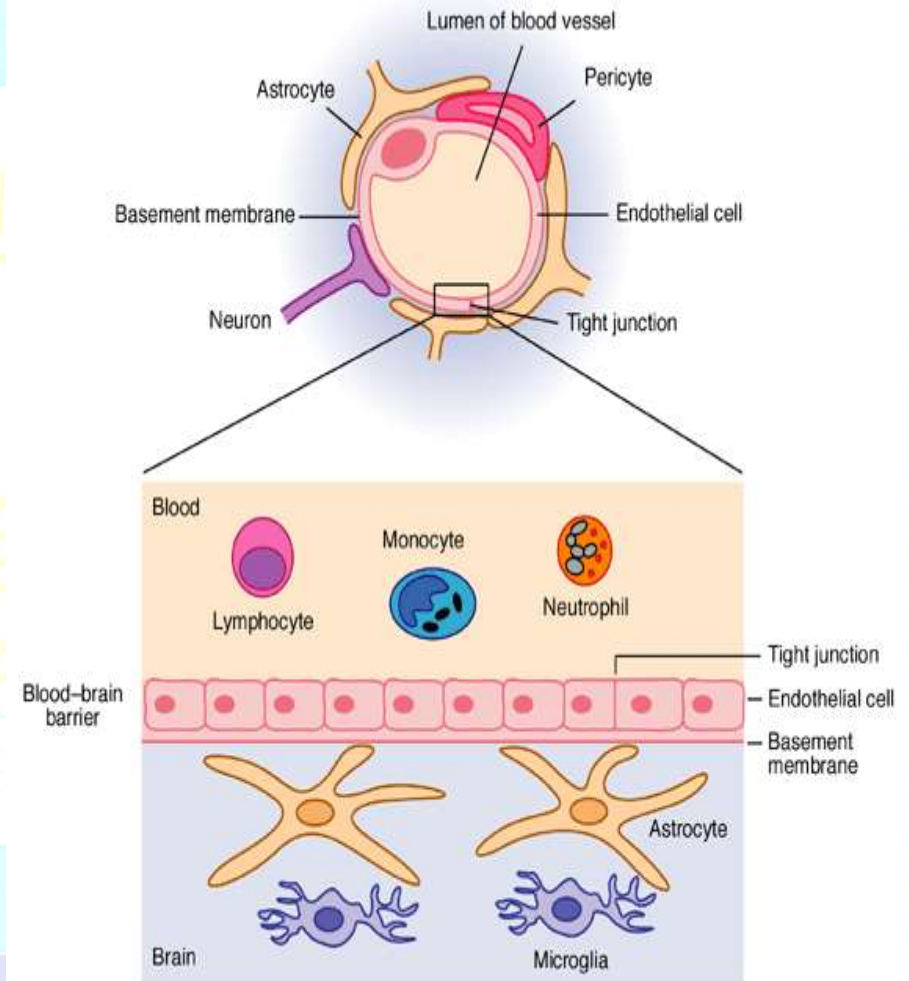


BLOOD-BRAIN BARRIER

- Endothelial cells connected by tight junctions
- Astrocytic endfeet surround brain microvessels.
- Only lipid soluble compounds diffuse through
- Water and ions: channels in apical and basal cell membranes
- Larger polar substances: by active transport
- Transport is selective between blood and brain, and between blood and spinal cord
- Blood-brain barrier is intact throughout except:
 1. Portions of hypothalamus (hormones)
 2. Capillaries in posterior pituitary
 3. Capillaries in pineal gland
 4. Capillaries of choroid plexuses

BLOOD-BRAIN BARRIER

- Collectively, the blood vessels within the brain have a very large surface area that promotes the exchange of oxygen, carbon dioxide, amino acids, and sugars between blood and brain.
- The blood-CSF barrier is formed by active transport from the blood vessels to the brain- Epithelial cells of joined by tight junctions, form a continuous layer that selectively permits the passage of some substances but not others.



The blood-brain barrier (BBB)

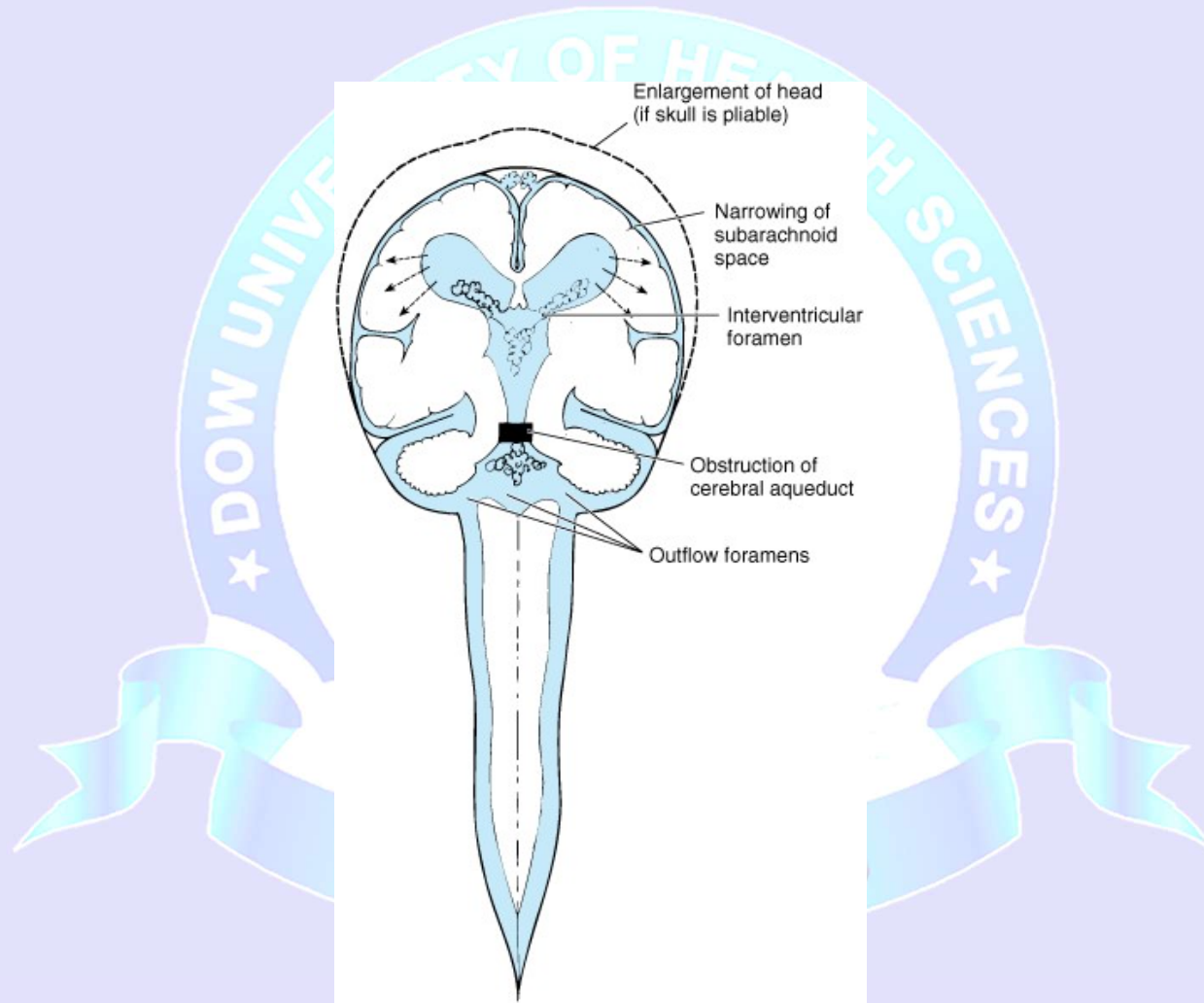
APPLIED ANATOMY; HYDROCEPHALUS

- **Hydrocephalus** also known as "water on the brain",
- is a medical condition in which there is an abnormal accumulation of cerebrospinal fluid (CSF) in the ventricles, or cavities, of the brain
- is caused by impaired cerebrospinal fluid (CSF) production, flow or reabsorption.
- The most common cause of hydrocephalus is a flow obstruction, hindering the free passage of cerebrospinal fluid through the ventricular system and subarachnoid space (e.g. stenosis of the cerebral aqueduct, obstruction of the interventricular foraminae - **foramen of Monro**). This can be secondary to tumors, hemorrhages, infections or congenital malformations. It can also be caused by overproduction of cerebrospinal fluid (relative obstruction).

TYPES OF HYDROCEPHALUS

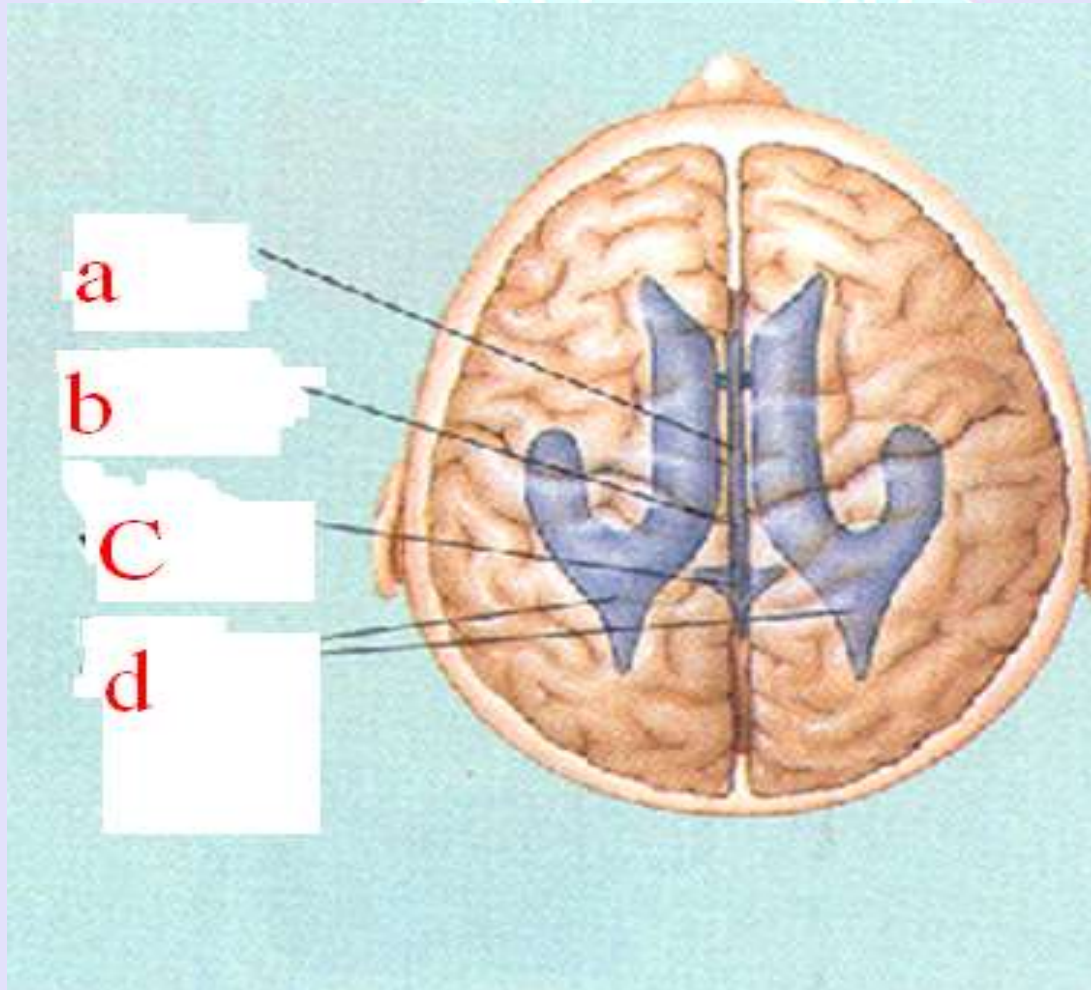
- ***OBSTRUCTIVE OR NON-COMMUNICATING***
(OBSTRUCTION WITHIN THE VENTRICULAR SYSTEM)
is caused by a CSF-flow obstruction ultimately preventing CSF from flowing into the subarachnoid space (either due to external compression or intraventricular mass lesions).
- ***NON OBSTRUCTIVE OR COMMUNICATING***
(MALFUNCTION OF ARACHNOID VILLI)
is caused by impaired cerebrospinal fluid resorption in the absence of any CSF-flow obstruction between the ventricles and subarachnoid space.

NON-COMMUNICATING **HYDROCEPHALUS**



SELF ASSESSMENT

Q1.Name the ventricles marked as a,b,c and d.



SELF ASSESSMENT

- Q2. Give the location of fourth ventricle.
- Q3. Name the structure forming the floor of fourth ventricle.
- Q4. What is the normal flow of CSF?
- Q5. What is the clinical importance of CSF flow in ventricular system?
- Q6. Name and define types of hydrocephalus.